

FUTURE IS NUCLEAR

► **India is already** cooperating with the US and France on newer technologies like small modular reactors (SMR) and advanced modular reactors. Nearly 18 years ago, India inked a historic nuclear deal with the US, ending India's nuclear apartheid. However, it remained a non-starter due to operational hurdles.

Recently, a breakthrough was achieved when the US administration cleared decks for US-based Holtec International's SMR technology to three Indian firms, namely Holtec Asia, Tata Consulting Engineers Ltd and Larsen & Toubro Ltd.

India is also working on setting up five indigenous

SMRs by 2033.

Globally, there are 80 SMR designs under various phases of research, development and approval. SMR is essentially a term used for a nuclear power reactor which has a capacity of up to 300 MW and employs a modular design.

And what makes SMRs a scalable technology? There are modules which are prepared at the manufacturing base of the respective industries and are brought to the site for assembly. The exclusion zone of SMRs is not beyond the plant boundary and can be of a capacity as small as 55 MW, effectively reducing the time required

to set up a project. It is similar to a plug-and-play setup, designed to work immediately after being connected, without needing any complex installation or configuration. Bharat Small Modular Reactor (BSMR) is the indigenous SMR that India is developing.

The concept design of the lead unit of BSMR 200 MW has been completed, which includes deciding on the sizing of the nuclear reactor along with the primary heat transport system.

The demonstration unit of BSMR 200 is expected to be completed in six years after financial approval. Plant commissioning, followed by regular operation, will be feasible at the end of the 7th year. The estimated cost of the lead unit is ₹5,700 crore.

70 |

der state control due to security and safety issues.

The industry is also awaiting the finer details of the amendments that must be cleared in the Parliament. The Parliamentary Committee also observed that ensuring strict compliance with nuclear safety protocols, waste management regulations, and radiation protection measures is crucial to avoid potential hazards.

"We need to wait for things (amendment) to happen and then take a call. What I can tell you is that we are fully geared up for this transition," says Sinha.

Meanwhile, a task force has been constituted in the Department of Atomic Energy to study the amendments required in the Atomic Energy Act. It will also be looking into various aspects like whether private players can build, own, and operate nuclear power plants, nuclear safety, security, safeguards, fuel procurement, fabrication, waste management and reprocessing of used fuel. In addition, a separate task force

is also looking into the Civil Liability for Nuclear Damage Act to address the concerns raised by the private suppliers. The liability Act outlines the legal framework for handling liability in the event of a nuclear accident and channelises operators' liability to equipment suppliers.

Enhanced safety features are identified for incorporation in nuclear power plants deploying the BSR. These features are expected to reduce the mandatory exclusion zone radius to about 500m from the existing one km. The Parliamentary Committee suggested that a clear contractual framework should be developed, ensuring transparent agreements that prioritise safety, security, and accountability in nuclear energy projects.

For now, the role of private players is limited to smaller reactors but there is a possibility for industry being involved in higher capacity reactors in the years to come. **BT**

@richajourno